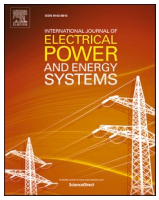
Electrical Power and Energy Systems 161 (2024) 110208



Contents lists available at ScienceDirect

International Journal of Electrical Power and Energy Systems

journal homepage: www.elsevier.com/locate/ijepes





Internal clock errors in synchrophasor ambient data: Effects, detection, and a posteriori estimation-based correction

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ARTICLE INFO

Keywords:
Bicoherence
Clock Error
Spectral Analysis

ABSTRACT

Synchrophasor data from real world power systems are exposed to numerous adverse cyber-and-physical operating conditions that can negatively impact their fitness for use in PMU applications. Although the most significant impairments such as GPS loss, instrument transformer failure, data drop-off, etc., have been studied in the literature, there exist other more complex nuisances that impact PMU data quality. This is the case of differences in how clock synchronization, time disciplining, and phasor estimation is performed by different PMU vendors, which have important implications on the PMUs' data fitness for use in PMU applications. In particular, ambient data applications, which have increasingly become a focus for tracking grid performance indicators, are extremely sensitive to periodic clock errors. Using synchrophasor and waveform data from Dominion Energy's power system, this article provides an in-depth analysis of the effect of seemingly minor device clock errors on the measurement signals' content. Finally, a method to distinguish clock errors from normal system dynamics and to correct them is proposed.

1. Introduction

The increased use of power electronic-based interfaces such as those in renewable energy sources and FACTS devices, has resulted in unprecedented dynamic performance challenges, particularly localized issues caused by poorly tuned controllers [1,2]. This is made worse by the lack of transparent, portable, and inter-operable dynamic models that makes it difficult for utilities to perform conventional physics-based model-based analysis, leaving measurements as the only source from where to derive actionable information. Because a typical power system mostly operates under ambient conditions characterized by small and persistent random perturbations [3], taking a proactive approach to controller performance monitoring requires the development of techniques capable of extracting and analyzing the dynamics of the underlying system purely from measurement data [1,2]. In this regard, frequency domain techniques [4] have proven to be effective due to the fact that the dynamic characteristics under ambient conditions resemble those of a linear stochastic system.

Working with real-world ambient data is difficult since the signal power containing relevant dynamics is far lower than what is seen during events (i.e., transients or large disturbances). As a result, even small discrepancies with IEEE Std. for Synchrophasor Measurements for Power Systems (IEEE C37.118.1-2011) [5] can have a significant impact on the ambient response observed in the data and, as a result, on data-driven applications.

For example, discrepancies in frequency estimation techniques inside PMUs of different suppliers [6] will greatly impact inertia monitoring [7].

Concerns related to time stamp corruption are the most challenging to address. The IEEE C37.118 std defines a Total Vector Error (TVE) limit of 1 %, which is equivalent to a phase angle error of 0.5730 (degrees) or a time synchronization inaccuracy of 21.6 μs at 60 Hz. Synchronization issues between different PMU devices result from discrepancies in their internal clocks and their disciplining, that are responsible for generating time stamps for the data. Here, the GPS time stamps arriving once a second provide a reference to these devices to discipline their internal clocks. How this is implemented differs from vendor to vendor. Figs. 1 and 2 demonstrate two relevant ways in which clock errors manifest in PMU phase angles in real-world field devices. Here, PMU A and B are from two different vendors and are within the same substation. PMU B occasionally (every 10–15 min.) disciplines its time stamps while A disciplines it precisely once a second. As a result, PMU B's phase angle

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https://doi.org/10.1016/j.ijepes.2024.110208
Received 13 February 2024; Received in revised form 30 June 2024; Accepted 27 August 2024
Available online 4 September 2024
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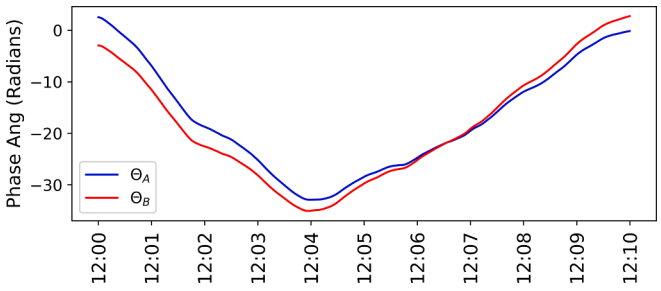


Fig. 1. Phase Angle Drifts from Loss of Synchronism.

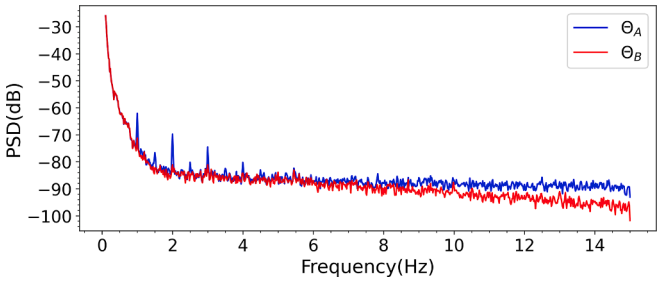


Fig. 2. Spectral Spikes from Periodic Discipling.

drifts w.r.t PMU A in Fig. 1. On the other hand, Fig. 2 shows the "detrended" phase angle plots in the frequency domain. Here, PMU A's angle has spectral spikes at 1 Hz and its harmonics [8], resulting from once a second discipling, which PMU B does not have. While time disciplining is the right thing to do in order to keep time stamp errors bounded, if not implemented properly, it poses a major issue for oscillation monitoring applications when done periodically, i.e., their spectrum may resembles low amplitude oscillations such as those shown in Fig. 2 between 0.5 – 4 Hz.

There are two general categories of timing errors. The first was first discussed in [9], which is termed as time skew and is the focus of the present work. The second is another family of time stamp errors resulting from GPS spoofing attacks [10,11]. What sets apart GPS spoofing attacks from the internal clock issues are that the latter happen upstream at the measurement device level. Furthermore, dealing with the GPS spoofing issues requires leveraging wide area system information, usually going through a state estimation step [12], unlike the device synchronization issues that require minimal "systems" information to be address.

The existing body of work focuses on the second category of timing issues, typically resulting in angle drifts. In practice, these usually result from events like loss of GPS signal [10,13,14], which is usually localized. Therefore, their detection and correction are based on finding a reference angle (most other locations on the grid) and estimating the drift in corrupted phase angle with respect to it. The authors in [15] discuss a related but inconspicuous issue with a constant time-stamp error. Its distinguishing property is a high correlation between phase angle and frequency, which is exploited to detect them followed by a similar correction to above. Finally, the second category of the more inconspicuous synchronization issues resulting from periodic disciplining are discussed in [9]. The treatment of the topic is limited in assessing the artifacts that the periodic clock errors create in frequency domain. The authors further propose a Kalman filtering based approach to correct them, given a reference angle. Furthermore, unlike the previous cases, *in our experience working with real-world synchrophasor data with a widespread deployment of PMUs, finding a reference angle free from these periodic errors with acceptable quality to be fit for ambient data applications is nearly impossible in practice*. In this paper, we will show that under a weak assumption of underlying power system stability, certain desirable phase angle signal properties allow us to detect and estimate these errors without needing a reference.

This builds upon our previous investigation of this issue in [16]. The contributions are as follows:

1. A theoretical analysis of the effect of periodic clock errors on phase angle spectral properties.
2. A method to distinguish between modal dynamics and periodic error using the phase angle bispectrum of ambient data.
3. An asymptotically unbiased estimator of periodic clock errors locally for stable ambient dynamics.

This paper is organized as follows. Section 2 of this paper gives a brief background in signal processing techniques used in our work. Next, Section 3 provides a brief description of the source of periodic clock errors, as well as a theoretical analysis of the problem using a combination of real measurements and synthetic signals. Section 4 proposes an approach to distinguish periodic clock errors from actual power system dynamics by exploiting bicoherence and an approach to estimate them. Finally, Section 5 demonstrates the effectiveness of our proposed work on synthetic and real-world data.

2. Spectral analysis background

This section will introduce some key signal processing techniques used in our work.

2.1. Power spectral density (PSD)

The PSD, denoted by $S(f)$, gives the signal's power distributed over frequency. For the stochastic process $x(t)$,

$$\int S(f)df = E\left(\lim_{T\to\infty}\int_0^T \frac{|x(t)|^2 dt}{T}\right) \tag{1}$$

where, $E(\bullet)$ is the expectation operator. Now, for ergodic processes [17], $\lim_{T\to\infty}\frac{1}{T}\int_0^T h(x(t))dt \to E(h(x(t)))$. Consequently, for a zero-mean wide-sense stationary ergodic process, $S(\omega)$ is,

$$S(f) = \int R(\tau)e^{-j2\pi f\tau}d\tau = \lim_{T\to\infty}\frac{E\left(|X(f)|^2\right)}{T} \tag{2}$$

where, $R(l) = E(x(t)x(t+l))$ is the autocorrelation function. In power system ambient measurements, there is uncertainty from driving noise as well as measurement noise, which needs to be accounted for when estimating PSD from measurement data [3]. Furthermore, in practice, we only have a unique response of the system (making realization of $E()$ difficult in practice) and, in addition, a finite data window. Therefore, to obtain a low variance estimate of the PSD, we use Welch's method [4] in this work, which operates by dividing the data window into parts and averaging spectral estimates over them, thus sacrificing frequency resolution for reduced variance.

2.2. Bispectrum and bicoherence

Higher-order moments are natural generalizations of the autocorrelation. A general p^{th} order cumulant $c_p(k_1,\cdots k_{p-1})$ (second order is autocorrelation) is defined as,

$$c_p(k_1,\cdots k_{p-1}) = E(x(t)x(t+k_1)\cdots x(t+k_{p-1})) \tag{3}$$

The Bispectrum [18], given in (4), is defined as the Fourier transform of the 3rd order cumulant and can yield insight into quadratic coupling between different frequency terms in a signal, usually resulting from non-linearities.

$$B(f_1,f_2)=\iint c_3(t_1,t_2)e^{-j2\pi(f_1t_1+f_2t_2)}dt_1dt_2=\lim_{T\to\infty}\frac{E(X(f_1)X(f_2)X^*(f_1+f_2))}{T} \tag{4}$$

where, $X(f)=\int x(t)e^{-j2\pi ft}dt$ is the fourier transform of $x(t)$ evaluated at ω. Finally, the Bicoherence $b(f_1,f_2)\in[0,1]$, is a measure of phase coupling between frequencies f_1,f_2 and f_1+f_2, and is given by,

$$b^2(f_1,f_2)=\frac{|E(X(f_1)X(f_2)X^*(f_1+f_2))|^2}{|E(X(f_1)X(f_2))|^2|E(X(f_1+f_2))|^2}. \tag{5}$$

3. Clock issues

Fig. 3 shows atypical symmetric patterns in the time–frequency plane (17:00–23:00) of the phase angle derivative obtained from data of a PMU at the 500 kV of a transformer within nuclear power plant. These patterns will be dissected in this work, as they were eventually found to be pertinent to the periodic clock errors, thus expanding the previously proposed conjecture.

3.1. Abnormal spectra in real-world measurements

To understand Fig. 3, we first start with a 10-minute window at 8:00, a portion of the spectrogram with less 'activity'. The most prominent feature is exhibited by the voltage angle spectrum in Fig. 4, characterized by sharp peaks/spikes at 1 Hz and its multiples [8]. These are relatively small in the voltage scale spectrum and are discussed in the next section. Next, from Fig. 3, we analyse a 10 min. window at 19:00, which appears to be significantly different, most likely due to operating changes in the power system around 17:00. Comparing the voltage and corresponding current phase angles reported by the same device, in Fig. 5, we see that a unique feature of these spectra are two closely spaced peaks near 0.5 Hz. In addition, similar peaks to the left and right of the spectral spikes are present. Finally, observe how these spikes are not visible in the angle difference, implying that the underlying mechanism behind these affects the VI measurement pair equally.

3.2. Effect of clock errors in the time domain

Clock errors can be thought of as a mechanism that replaces actual time t by a spurious time τ. Let $\phi: t\to\tau$ represent the time stamp corruption mechanism and ψ be the inverse mapping (i.e., a mapping that recovers the true time stamp from a fictitious one). In this work, we assume that the clock error is small, i.e., $\tau=\phi(t)=t+\epsilon\phi_1(t)$ for some small ϵ, which in practice is necessary to satisfy the standard IEEE C37.118.1-2011 requirements [5]. Thus, this is a near-identity transform, for which an inverse ψ exists and has the form $t=\psi(\tau)=\tau+\epsilon\psi_1(\tau)$ as long as $\partial_t\phi(t)=1+\epsilon\partial_t\phi_1(t)\neq 0\,\forall t$ according to the inverse function theorem [19].

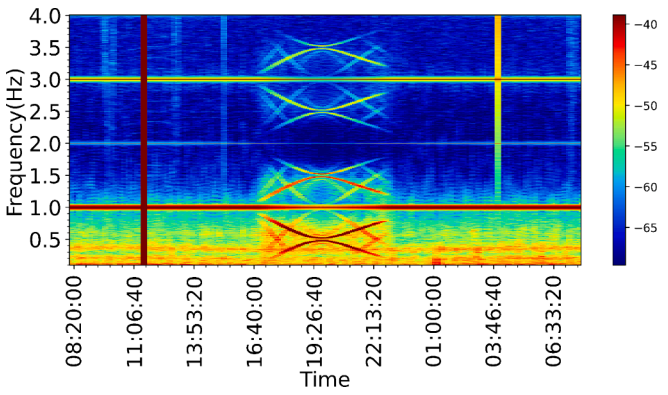


Fig. 3. Voltage Angle Derivative Spectrogram Showing Atypical Patterns.

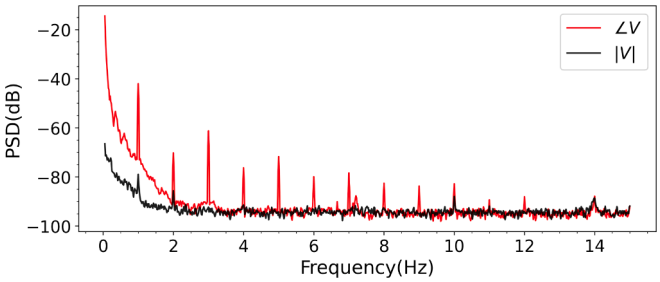


Fig. 4. Voltage Magnitude and Voltage Angle Spectrum.

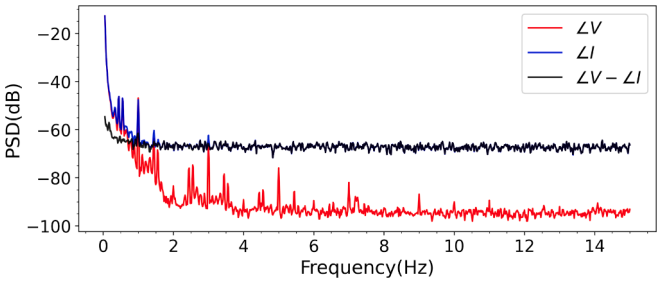


Fig. 5. PMU Voltage Angle and Current Angle Spectra.

Let us now examine the effect of such an error on the apparent system dynamics based on the measured data. Let the true system trajectory be denoted by $x(t)$. Note that, in practice, we only have access to the corrupt time stamp, τ, and not the true time stamp, t. Consequently, τ is the only time practically available and all calculations will be done w.r.t. τ in the rest of this work. Hence, at a given corrupted time τ, what we measure is $x(t)=x(\psi(\tau))$ instead of $x(\tau)$. Letting x be the states of an autonomous system, i.e., $\frac{dx(\tau)}{d\tau}=f(x(\tau))$, the perceived vector field would be $\frac{dx(\psi(\tau))}{d\tau}=\frac{d\psi}{d\tau}f(x(\psi(\tau)))$, i.e., an nonautonomous system.

Example 1. Let the original/correct system trajectory be exponentially decaying sinusoid $x(t)=e^{-0.1t}\cos(2\pi\times 0.2\times t)$, resembling impulse response of a system with a single damped mode at 0.2 Hz. Let the clock corruption be of the form $\phi(t)=2t\to\psi(\tau)=\frac{\tau}{2}$, i.e., the clock is significantly faster than the true time t and not disciplined. This is typical when GPS signal is lost and the internal clock drifts. The perceived dynamics are shown in Fig 6 and are equal to $x(\psi(\tau))=e^{-0.05\tau}\cos(2\pi 0.1\tau)$. As it can be observed, not only is the frequency modified, but the apparent damping is also lower than that of the original/correct signal.

3.3. Explaining atypical spectral symmetries in ambient data

For convenience, in this section, $\epsilon\psi_1(\tau)=t(\tau)-\tau$ will be denoted by $\Delta t(\tau)$. Before proceeding, we need to understand the clock error's structure $\Delta t(\tau)$, as it exists in real-world settings. Most of the timing in substations today is generated by a satellite-synchronized clock. This clock provides timing to many devices in the substation and the most

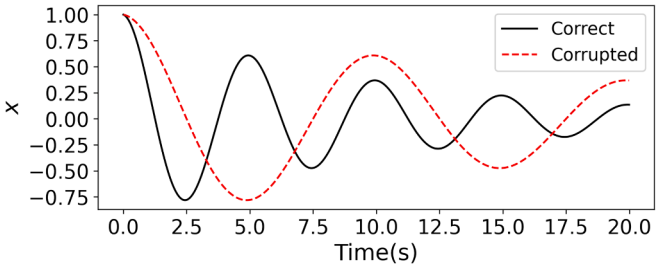


Fig. 6. Example 1- True vs Corrupted Dynamics.

common method of distributing timing signal is via an IRIG-B signal. Substation devices, e.g., digital relays and/or PMUs, decode this once per second time signal and use it to synchronize/discipline their own internal time circuit periodically. This circuit is responsible for generating sub-second time stamps for the measurements. This periodic disciplining results in time stamp errors $(\phi(t)-t)$ having a periodicity with a period T_0(ideally 1*sec.*), i.e., $\phi_1(t) = \phi_1(t+nT_0)\forall n$. Thus, $\phi(t+nT_0)-\phi(t) = (t+nT_0)-t = nT_0 \forall t, n$. Taking the inverse, $\psi()$, shows that the same periodicity of T_0 exists in $\psi_1(\tau)$ and therefore $\Delta t(\tau) = \sum_{k\in Z} a_k e^{j2\pi k f_0 \tau}$, where the fundamental frequency of the clock error is $f_0 = \frac{1}{T_0} Hz$ and $a_{-k} = a_k^*$. These periodic clock error components will be referred to as *"clock modes"* herein to distinguish from natural system modes. Let us also define the set of all clock frequencies as $A_{f_0} = \{kf_0 \forall k \in Z\}$.

To study the effect of this periodicity, a first order Taylor expansion of the measured signal $x(\psi(\tau))$ around τ is taken,

$$x_\psi(\tau) = x(\tau) + \dot{x}(\tau)\Delta t(\tau). \tag{6}$$

Next, power system ambient dynamics can be modeled as a zero mean wide sense stationary process for which $E(X(f)X^*(f_1)) = S_x(f)\delta(f-f_1)$ [4], where $S_x(f)$ denotes the PSD. To understand the measured signal spectrum, denote by $S_{x_\psi}(f)$, we substitute $\Delta t(\tau) = \sum_{k\in Z} a_k e^{j2\pi f_0 \tau}$ in the expression above. Here, the Fourier transform of $\Delta t(\tau)$, i.e., $\mathscr{F}\{\Delta t\}(f) = \sum_{k\in Z} a_k \delta(f-kf_0)$ where $\mathscr{F}\{\}$ is the Fourier transform operator. Taking a Fourier transform of the above equation,

$$\begin{aligned} X_\psi(f) &= X(f) + \sum_{k\in Z} a_k \mathscr{F}\{\dot{x}\}(f) * \delta(f-kf_0) \\ &= X(f) + \sum_{k\in Z} a_k \mathscr{F}\{\dot{x}\}(f-kf_0) \\ &= (1 + ja_0 2\pi f)X(f) + \sum_{k\neq 0\in Z} a_k j2\pi (f-kf_0) X(f-kf_0) \end{aligned} \tag{7}$$

where $X(f) = \mathscr{F}\{x\}(f)$ and $X_\psi(f) = \mathscr{F}\{x_\psi\}(f)$ and * is the convolution operator. For a wide sense stationary signal, $x(\tau)$, we have the property that that $E(X(f_1)X^*(f_2 \neq f_1)) = 0$, thus, the power spectral density can be computed as,

$$\begin{aligned} S_{x_\psi}(f) &= \lim_{T\to\infty} \frac{E\left(|X_\psi(f)|^2\right)}{T} \\ &= S_x |1 + ja_0 2\pi f|^2 + \sum_{k\in Z} |a_k|^2 (2\pi)^2 (f-kf_0)^2 S_x(f-kf_0) \end{aligned} \tag{8}$$

From Equation (8), it is important to note that the perceived spectrum contains a scaled version of the original spectrum by a factor of $|1 + ja_0 2\pi f|^2$. As a result, while $S_x(f)$ and $S_{x\psi}(f)$ are nearly the same at low frequencies, their differences become noticeable at higher frequencies. This can be seen in Fig. 2 where Θ_B, which does not have a periodic correction has its spectrum roll off at higher frequencies while $\Theta_A's$ does not. In addition, observe how $S_x(f)$ is present at precisely all frequencies of the form $nf_0 + f, n \in Z$ in $S_{x_\psi}(f)$. This helps, in part, to explain the spectral symmetries seen in Fig. 3. This duplication could particularly be deceptive in synchrophasor-based dynamic monitoring and assessment, specially mode estimation applications [20], and control[21] . For example, assume we have a poorly damped mode at some $f = f_{mode}$, which results in a relatively large value of $S_x(f_{mode})$. Multiple scaled copies of these dynamics will appear in the neighbourhood of all clock frequencies, $A_{f_0} = \{kf_0 \forall k \in Z\}$, thereby requiring a much higher order model to characterize the dynamics observed. This is illustrated in the next example. Furthermore, it will be picked up by various controllers [22], that could make them attempt to suppress it, and in the process, injecting actual periodic input to the system.

Example 2. Let $x(t)$ be the response of a second order stochastic linear model. Here, $x(t)$ is similar to typical ambient dynamics with a single mode at 0.2 Hz with 1% damping. Assume the corrupted clock $\tau = t + \epsilon \frac{1+\cos(2\pi\times 1\times t)}{2}$ and consider two different values of $\epsilon = 10^{-1}$ and $\epsilon = 2\times 10^{-1}$. The power spectral density of $x(t)$ is shown in Fig. 7, where it can be observed that the smaller the clock errors, the fewer the visible clock harmonics.

Furthermore, in Fig. 7, it can be seen that the spectral spikes at clock frequencies A_{f_0} are themselves absent (dashed grey line) while their interaction with system dynamics (particularly the mode at 0.2 Hz) are present in the form of spectral peaks to the left and right of clock frequencies. This is different from what was observed in the actual real-world data shown in Fig. 5, which will be analysed in detail in later sections.

In addition to the results in Equation (8), we estimate $E(X_\psi(f_1)X_\psi^*(f_2 \neq f_1))$, which will be useful in following sections particularly for detection,

$$\begin{aligned} E(X_\psi(f_1)X_\psi^*(f_2 \neq f_1)) = &\sum_{k\in Z} a_k^* E(X(f_1)\mathscr{F}\{\dot{x}\}(-f_2 \\ &+ kf_0)) + \sum_{k\in Z} a_k E(X(-f_2)\mathscr{F}\{\dot{x}\}(f_1 \\ &- kf_0)) + \sum_{k_1\in Z}\sum_{k_2\in Z} a_{k_1} a_{k_2}^* E(\mathscr{F}\{\dot{x}\}(f_1 \\ &+ k_1 f_0)\mathscr{F}\{\dot{x}\}(-f_2 + k_2 f_0)). \end{aligned} \tag{9}$$

Observe that Equation (9) is zero $\forall (f_1, f_2)$ for which $f_1 - f_2 \notin A_{f_0}$, i.e., not separated by an integer multiple of clock fundamental frequency f_0.

3.4. Effect of signal trend on PMU spectra and waveform data

Next, we aim to understand the effect of trends in $x(t)$ on the perceived spectrum S_{x_ψ}. From Equation (8), it can be observed that the clock mode at a_k can only appear at its own frequency kf_0 through $\dot{x}(\tau)$'s spectral content at $kf_0 - kf_0 = 0$ Hz. For a mean stationary process $x(\tau)$, this is not possible. This explains the absence of any spectral spikes at the clock frequencies in Example 2. However, if we add a trend term to the original ambient data signal, $x(\tau)$, e.g., $y(\tau) = b_0 + b_1\tau + x(\tau)$, the corrupted signal then becomes $y_\psi(\tau) = b_0 + b_1\tau + b_1\Delta t(\tau) + x_\psi(\tau)$. Now, the PSD is defined for stationary processes (at least in mean) and therefore, the signal will typically be detrended before estimating it, i.e.,

$$detr(y_\psi(\tau)) = \underbrace{b_1\Delta t(\tau)}_{trend\times clock} + \underbrace{x_\psi(\tau)}_{\substack{corrupted ambient \\ dynamics}}. \tag{10}$$

Note here that detrending does not remove the trend's interaction terms with the clock error $\Delta t(\tau)$ since that interaction is sinusoidal. Finally, the detrended clock-corrupted signal in frequency domain becomes,

$$Y_\psi(f) = b_1 \sum_{k\neq 0\in Z} a_k \delta(f-kf_0) + X_\psi(f) \tag{11}$$

where $X_\psi(f)$ defined by Equation (7). This will help us fully explain the real-world observations, as well as why we see prominent clock modes in only phase angles.

Power system dynamics around 60 Hz can be modeled as a

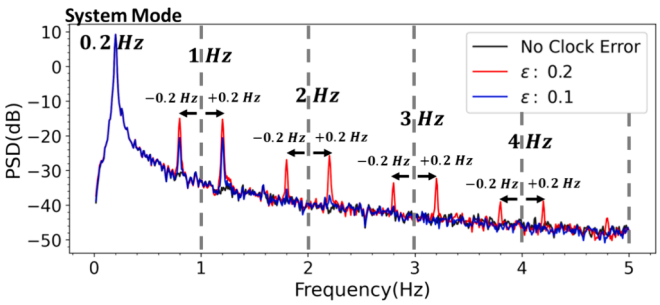


Fig. 7. Example 2 – Corrupted Signal Spectrum.

modulated 60 Hz wave [1],

$$x_{waveform}(t) = A(t)e^{i\theta(t)} \\ \theta(t) = 2\pi f_c t + \rho(t) \tag{12}$$

where, $A(t)$ and $\theta(t)$ are amplitude and phase angle of the waveform, respectively. The phase angle is comprised by a nearly constant drift term from the carrier frequency $f_c \approx 60$ Hz and the remaining system dynamics $\rho(t)$. Synchrophasor estimates [23] from PMUs aim to extract the amplitude $A(t)$ and $\theta(t) - 2\pi \times 60 \times t = \rho(t) + 2\pi(f_c - 60) \times t$. To illustrate, consider Fig. 8, which shows the spectrum of the waveform data for the previously analysed window in Fig. 5.

In Fig. 8 the carrier frequency around 60 Hz has the highest peak. Meanwhile, system dynamics are visible as side bands (to the left and right) of the carrier' peak. The frequencies of these sidebands are at $f_c \pm kf_{mode} \forall k$ where f_{mode} is the corresponding oscillation/mode frequency. In Fig. 8, the most dominant sidebands are seen at $f_{mode} = 60 \pm 1$ Hz, which correspond to the clock modes. The same clock modes can be seen around the second and higher harmonics of the 60 Hz carrier. This confirms that these are not artifacts of the synchrophasor estimation algorithm, instead, they a result of a signal whose time stamp has been corrupted during its acquisition at the waveform stage, as expected from internal clock errors. As a result, the large trend term $\approx 2\pi \times 60 \times t$ in the phase angle $\theta(t)$ needs to be accounted for when analyzing the effect of clock errors. Based on our previous discussion in this section, the trends are responsible for placing clock modes at their own frequencies (in the form of spectral spikes), i.e., the larger the trend slope $|b_1|$, the more pronounced its effect on the final spectrum. Note that in general, slow trends are there in all measurement signals resulting from changing load and generation throughout the day_[24]. However, trends in voltage and current magnitude are negligible compared to the $2\pi \times 60$ (see Equation (12)) in phase angles, which explains prominent spectral spikes in only angles, as confirmed by Fig. 4.

4. Detection and estimation

In this section, in accordance with the norm in modal analysis [24], power system ambient dynamics will be modeled as output of stable linear system (with impulse response $g(t)$ driven by stationary gaussian white noise $u(t)$ $N(0,1)$. Based on our previous discussions, we add a DC component and a trend term to yield the following ambient response,

$$y(\tau) = \underbrace{b_0 + b_1\tau}_{DC+trend} + \underbrace{\int g(\tau - \eta)u(\eta)d\eta}_{x(\tau)}. \tag{13}$$

Note that here, the dependence of ambient dynamics on the trend is ignored, as commonly assumed in modal analysis applications by assuming stationarity of spectrum of ambient dynamics [25]. The clock-corrupted measurements can then be modeled as,

$$y_\psi(\tau) = b_0 + b_1\tau + b_1\Delta t(\tau) + \int \underbrace{(g(\tau - \eta) + \partial_\tau g(\tau - \eta)\Delta t(\tau))}_{=h(\tau,\eta)} u(\eta)d\eta. \tag{14}$$

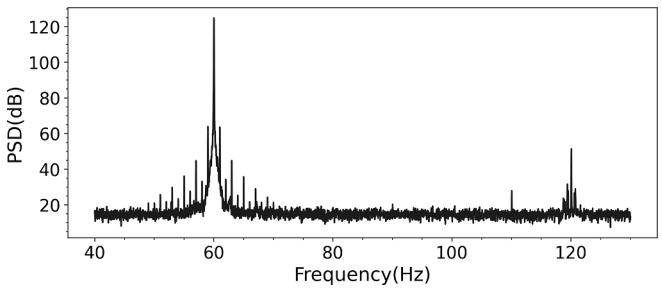


Fig. 8. Voltage Waveform Data Spectrum.

Note from Equation (14) how the clock-corrupted dynamics can no longer be modeled as output of a linear time-invariant system due to having a time-varying impulse response of the form $h(t_1, t_2) = g(t_1 - t_2) + \partial_{t_1} g(t_1 - t_2)\Delta t(t_1)$ along with an oscillatory component $b_1\Delta t(\tau)$. Here, we also define $G(f) = \mathscr{F}\{g(\tau)\}(f)$ and $U(f) = \mathscr{F}\{u(\tau)\}(f)$ with $X(f) = G(f)U(f)$.

4.1. Distinguishing clock errors from true system dynamics

While second order spectra (PSD) show the effects of clock error, the characteristics of this spectra are such as that arising from the power system's true dynamics. One such example is voltage sags [26], which result in a similar spectral characteristic. Therefore, we need to determine how these clock errors affect the signal and create cross frequency coupling terms as seen in the previous section.

Firstly, the Bispectrum of (detrended) ambient data with no clock issues ($Y(f) = G(f)U(f)$) can be estimated as [27],

$$B_y(f_1, f_2) = \lim_{T\to\infty} \frac{1}{T} G(f_1)G(f_2)G^*(f_1 + f_2)\underbrace{E(U(f_1)U(f_2)U^*(f_1 + f_2))}_{=0(zeromean\ gaussianwhitenoise)}. \tag{15}$$

Next, we derive the expression for the Bipsectrum of a (detrended) clock-corrupted $y_\psi(\tau)$. For that, we start with $Y_\psi(f) = b_1\mathscr{F}\{\Delta t\}(f) + X_\psi(f)$ where $X_\psi(f)$ is from (7). Now, odd order moments of $u(t)$ are zero. Since all terms in $X_\psi(f)$ are linear in $U(f)$ when substituting $X(f) = G(f)U(f)$, the odd degree terms (in $X_\psi(f)$) in the Bispectrum are all 0. Therefore, retaining only even degree terms in $X_\psi(f)$ gives,

$$\begin{aligned} B_{y_\psi}(f_1, f_2) &= b_1\mathscr{F}\{\Delta t\}(f_1)E(X_\psi(f_2)X_\psi^*(f_1 + f_2)) \\ &+ b_1\mathscr{F}\{\Delta t\}(f_2)E(X_\psi(f_1)X_\psi^*(f_1 + f_2)) \\ &+ b_1\mathscr{F}^*\{\Delta t\}(f_1 + f_2)E(X_\psi(f_1)X_\psi(f_2)) \\ &+ b_1^3\mathscr{F}\{\Delta t\}(f_1)\mathscr{F}\{\Delta t\}(f_2)\mathscr{F}^*\{\Delta t\}(f_1 + f_2). \end{aligned} \tag{16}$$

Firstly, we see that without a trend ($b_1 = 0$), the Bispectrum is 0, which shows the importance of the underlying signal trend in exposing clock issues. Now, from Equation (9) and the fact that Δt has discrete spectrum, i.e., $\mathscr{F}(\Delta t)(f) = 0 \forall f \notin A_{f_0} = \{kf_0 \forall k \in Z\}$, the only possible non-zero Bispectrum terms are at frequency pairs in the set $\{(f_1, f_2) \in R^2 | f_1 \in A_{f_0} or f_2 \in A_{f_0} or f_1 + f_2 \in A_{f_0}\}$, yielding the following table,

To intuitively understand the table above, take the example of a mode at a non-clock frequency, i.e., $f_{mode} \notin A_{f_0}$. As seen before, we see its copies at $f_1 \pm f_{mode} \forall f_1 \in A_{f_0}$. Substituting $f_2 = f_{mode}$ in the table above tells us that this will make the first term in the Bispectrum non-zero, and therefore, it is possible to observe it. To summarize, the Bispectrum of a synchrophasor signal can help us distinguish clock created pseudo modes from real dynamics (where the Bispectrum will yield zero at all frequencies).

4.2. Estimation

Here, the internal clock error's fundamental frequency f_0 is assumed to be known beforehand. In addition, the clock error is assumed to be bounded, i.e., $\max_\tau(|\Delta t(\tau)|) = \Delta t_{max} < \infty$. The corrupted signal is given by $y_\psi(\tau)$ in (14). What makes this estimation problem challenging in practice are identifiability concerns posed by a lack of a reference and/or knowledge of the true system. However, assuming stable ambient dynamics (governed by impulse response g), which holds true in general, and a large trend ($|b_1|$) (such as $2\pi \times 60$ in the phase angle), it becomes possible to recover $\Delta t(\tau)$ from clock corrupted measurements $y_\psi(\tau)$. The estimation essentially involves: 1) estimating the trend in the signal, followed by 2) estimating the clock error $\Delta t(\tau)$ and 3) recovering the original/correct measurement.

4.2.1. Estimating the trend

Since $\Delta t(\tau)$ is multiplied with b_1, b_1 needs to be estimated as well. This can be posed as a linear regression problem,

$$Y(\tau) \approx X(\tau) \times \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} + e(\tau) \tag{17}$$

where, $X(\tau) = [1, \tau]$, $Y(\tau) = y_\psi(\tau)$ and $e(\tau) = b_1 \Delta t(\tau) + \int h(\tau, \eta) u(\eta) d\eta$. For a trajectory spanning T seconds, the ordinary least squares (OLS) estimates $\widehat{b}_0, \widehat{b}_1$ satisfy,

$$\begin{bmatrix} \widehat{b_0} - b_0 \\ \widehat{b_1} - b_1 \end{bmatrix} = \begin{bmatrix} 4 & -6 \\ -\frac{6}{T} & \frac{12}{T} \end{bmatrix} \times \begin{bmatrix} \frac{\int_0^T e(\tau) d\tau}{T} \\ \frac{\int_0^T \tau e(\tau) d\tau}{T^2} \end{bmatrix}. \tag{18}$$

Note that $e(\tau)$ in Equation (17) are not independent in τ, which violates a major assumption in OLS. Furthermore, $E(e(\tau)) = b_1 \Delta t(\tau) \neq 0$ and therefore, it is not possible to separate the DC component of $\Delta t(\tau)$, i.e., a_0 from the DC component of the trend, i.e., b_0. Next, we will show that $\widehat{b}_1$, obtained from OLS, asymptotically converges to b_1 as $T \to \infty$. To do so, we go through a series of lemma (with proofs in the Appendix),

Lemma 1. $\lim_{T\to\infty} \frac{\int_0^T e(\tau) d\tau}{T^2} \xrightarrow{p} \frac{b_1 a_0}{T} = 0$ where $\xrightarrow{p}$ denotes convergence in probability [17].

Lemma 2. $lim_{T\to\infty} \frac{\int_0^T \tau e(\tau) d\tau}{T^3} \xrightarrow{p} 0$

Theorem 1. $\lim_{T\to\infty} \widehat{b}_1 - b_1 \xrightarrow{p} 0$

Proof. From (18), lemma 1 and 2, $\lim_{T\to\infty} \widehat{b}_1 - b_1 = -6 \frac{\int_0^T e(\tau) d\tau}{T^2} + 12 \frac{\int_0^T \tau e(\tau) d\tau}{T^3} \xrightarrow{p} 0$.

Theorem 1 states that the OLS estimate $\widehat{b}_1$, i.e., the linear trend slope b_1, asymptotically converges to its true value. On the other hand, it can be seen from Equation (14) that b_0 is indistinguishable from $b_1 \Delta t(\tau)$'s DC component ($b_1 a_0$) and therefore cannot be estimated. This is not of concern for ambient data used in oscillation analysis because Equation (7) shows that a_0 does not "create" new "pseudo" modes above 0 Hz and therefore, it is not necessary to estimate it. Thus, the focus would be on estimating other components, i.e., components $a_{k>0}$ (since $a_{-k} = a_k^*$). Note that under the special case of the error $\Delta t(\tau)$ resetting to zero at the end of each correction cycle of $\frac{1}{f_0}$ seconds translates to the constraint $\Delta t\left(\frac{n}{f_0}\right) = a_0 + \sum_{k\neq 0} a_k$, from where we can estimate otherwise unidentifiable a_0 as $\widehat{a}_0 = -\sum_{k\neq 0} \widehat{a}_k$.

4.2.2 Estimating clock error

Finally, we propose the following estimator for the periodic components of $\Delta t(\tau)$,

$$\widehat{a}_k(T) = \frac{1}{\widehat{b}_1 T} \int_0^T (y_\psi(\tau) - \widehat{b}_1 \tau) e^{-jk2\pi f_0 \tau} d\tau \forall k \neq 0$$
$$\Delta \widehat{t}(\tau) = \sum_k \widehat{a}_k e^{j2\pi k f_0 \tau}. \tag{19}$$

Here, $\lim_{T\to\infty} \widehat{b}_1 - b_1 \xrightarrow{p} 0$, is proven in Theorem 1. Next, the bias and variance of $\widehat{a}_k$ for a finite data window spanning T (assuming T is an integer multiple of clock time-period $\frac{1}{f_0}$), given the trend b_1, can be shown to be (see proof in Appendix),

$$E(\widehat{a}_k(T) | \widehat{b}_1 = b_1) = a_k \tag{20}$$

$$VAR(\widehat{a}_k(T) | \widehat{b}_1 = b_1) \approx \frac{1}{b_1^2} \int_0^T \int_0^T \frac{1}{T^2} R(\tau - \tau_1) e^{-jk2\pi f_0 (\tau - \tau_1)} d\tau d\tau_1 \tag{21}$$

where, $R(t) = \int g(\eta) g(\eta + t) d\eta$. Thus, the proposed estimator $\widehat{a}_k$ is unbiased. Furthermore, $VAR(\widehat{a}_k(T))$ is dependent on the underlying system's impulse response at $f = kf_0$ and a factor $\frac{1}{b_1^2}$. Consequently, for signals with a sufficiently large linear trend $|b_1|$, such as in the case of phase angles, the estimator is of low variance. In situations where the trend $|b_1|$ is small, a longer window T is required to obtain a reduced estimator variance. For a linear system driven by a Gaussian white noise input with impulse response $g(t)$, $R(t)$ is its autocorrelation function and consequently, its Fourier transform is the power spectral density (PSD), as presented in Section 2.1 Thus, longer windows will be needed for a clock frequency kf_0 if the underlying system has a poorly damped mode at/in the neighborhood of that frequency, which results in a high PSD value. In the rare case that an undamped system mode appears at one or more clock frequencies, the estimator may need an infeasibly long window.

4.2.3. Recovering correct measurement from the clock error estimate

The true time stamp estimate using the estimated clock error $\Delta \widehat{t}(\tau)$ is given as $\widehat{t}(\tau) = \tau + \Delta \widehat{t}(\tau)$. To recover the true signal $\widehat{y}(\tau)$ from the clock-corrupted signal, $y_\psi(\tau)$, any standard interpolation function can be used. Such method is denoted herein by *INTRP*(), which takes data points in the form of time array t_{data} and value array x_{data} and returns the interpolated values y_{new} at t_{new} i.e. $y_{new} = INTRP(t_{new}, t_{data}, x_{data})$,

$$\widehat{y}(\tau) = INTRP(\tau, \widehat{t}(\tau), y_\psi(\tau)) \tag{22}$$

The results in the preceding three sub-sections leads to an overall algorithm for detection and correction approach, which is detailed below.

Overall Algorithm: Correcting Periodic Clock Error Corrupted Signal from Stable Ambient Conditions

Input: Corrupted measurements $y_\psi(\tau)$ with time stamp τ for a window spanning T seconds in ambient conditions.

Detection: Obtain the PSD of $y_\psi(\tau)$, $S_{y_\psi}(f)$, using any approach (e.g., Welch's method [4]). Determine if several spectral spikes (narrow band line spectra) separated by some fixed frequency f_0 exist in the PSD. In addition, determine if any other spectral peaks (representing mode dynamics) exist at frequencies f_{mode}.

If present, estimate the Bicoherence $b^2(f_1, f_2)$. The presence of significant > 0.5 bicoherence values between spectral spike frequencies $kf_0 \forall k \in Z^+$ and modal frequencies in the set f_{mode} points to a potential periodic clock error issue.

If present, proceed to the correction step.

Correction

Estimate the trend coefficient b_1 by solving the following least squares problem.

$y_\psi(\tau) = b_0 + b_1 \tau$

Using the estimate $\widehat{b}_1$, estimate the clock mode coeffcients a_k as,

$\widehat{a}_k = \frac{1}{\widehat{b}_1 T} \int_0^T (y_\psi(\tau) - \widehat{b}_1 \tau) e^{-jk2\pi f_0 \tau} d\tau$

Finally, estimate the corrected signal $\widehat{y}(\tau)$ using interpolation as,

$\widehat{t}(\tau) = \tau + \sum_k a_k e^{j2\pi k f_0 \tau}$

$\widehat{y}(\tau) = INTRP(\tau, \widehat{t}(\tau), y_\psi(\tau))$

Output: $\widehat{y}(\tau)$

5. Results

5.1. Effect of trend component

This first example aims to illustrate the fundamental results presented in this paper along with illustrating the proposed algorithm. First, we study the effect of trend in the underlying signal and how that influences the detection and correction of clock errors. We use a randomly generated 10 min trajectory of the stochastic system from Example 2, we add three different trends with $b_1 = 0$, 0.1 and 1, respectively, and corrupt the signal it with a periodic clock error of the form $\tau = t + 0.2 \times \frac{1+\cos(2\pi \times 1 \times t)}{2}$. The time period is $T_0 = 1$ sec., with the clock frequencies $A_{f_0} = \{\cdots, -2, -1, 0, 1, 2 \cdots\} Hz$. The clock-corrupted signal's spectral estimates are plotted in Fig. 9.

It is evident that larger the trend, more prominent the presence of

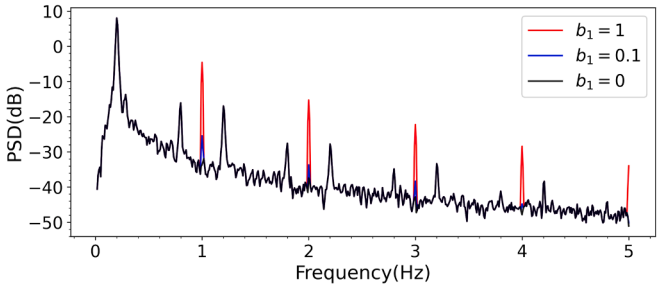


Fig. 9. Effect of a Linear Trend on PSD of Clock-Corrupted Signals.

clock modes at their own frequencies in the set A_{f_0}, which was analytically shown in (10). More importantly, one can observe the many new spectral peaks at $f \pm 0.2 \forall f \in A_{f_0}$, which results from the interaction of the 0.2 Hz system mode with the clock error frequencies. As discussed before, these are not true dynamics, which will be shown next through bicoherence.

The Bicoherence estimate for detection is plotted in Fig. 10. For the case where $b_1 = 1$, significant Bicoherence can be observed between the clock frequencies in the set A_{f_0} and the mode frequency at 0.2 Hz, which do not appear for the case of trend $b_1 = 0$. This confirms our analytical results from Table 1 and more importantly, shows that these are not true system dynamics.

Next, we randomly generate 100 realizations (trajectories) of the same system response, corrupt them each with the same clock error discussed above and estimate it for each using the proposed estimator from Section 4.2. Fig. 11 shows the estimates of Fourier coefficients a_k for $\Delta t(\tau) = \sum_k a_k e^{j2\pi k f_0 t}$ in color along with the true values denoted by crosses in the complex plane. The estimates were obtained using an estimation window of $T = 10$ min. Note that estimation uncertainty decreases for a larger underlying signal trend, which confirms our analysis in the previous section. This should make it possible to precisely estimate the clock errors in phase angles, which natively have a large trend value.

Finally, we use $\widehat{a}_k$ for one of the 100 signals (realizations) to correct the time stamps and consequently recover the true system dynamics in Fig. 12. As expected, correcting the time stamps not only removes the clock modes at clock frequencies but also the pseudo dynamics resulting from the interaction of 0.2 Hz mode with the clock error.

5.2. Mode at clock frequency

This example considers the extreme scenario of the presence of an underlying system mode at one of the clock frequencies and how its damping impacts our ability to estimate the clock error. To this end, we shift the 0.2 Hz mode from the previous case to one of the clock's frequencies, i.e., 2 Hz. In addition, we consider three different damping ratios of 1 %, 10 % and 50 % for this mode.

Table 1
Bispectrum Terms.

	1st Term	2nd Term	3rd Term	4th Term
$(f_1, f_2) \in A_{f_0}$	*	*	*	*
$f_1 \in A_{f_0}, f_2 \notin A_{f_0}$	*	0	0	0
$f_1 \notin A_{f_0}, f_2 \in A_{f_0}$	0	*	0	0
$f_1 \notin A_{f_0}, f_2 \notin A_{f_0}, f_1 + f_2 \in A_{f_0}$	0	0	*	0
$f_1 \notin A_{f_0}, f_2 \notin A_{f_0}, f_1 + f_2 \notin A_{f_0}$	0	0	0	0

Lowering the damping of a mode significantly increases the PSD of the response about mode's frequency (2 Hz in this case). However, from (8), the effect of periodic clock errors is placing this part of the signal at the frequencies in the set $\{f \pm f_{mode} \forall f \in A_{f_0}\}$. Therefore, as shown in Fig. 13, we see an increased signal power at many other frequencies, which as we know are pseudo dynamics. Now, as the damping is increased for this mode, the clock-corrupted signal becomes similar to that of pure clock modes (spectral spikes). Intuitively speaking, increased damping should make it easier to estimate the clock error correctly. To study the effect of mode damping on clock error estimation variance, we again generate 100 independent realizations of the system response for each damping level, estimate the clock error Fourier terms a_k using an estimation window of $T = 10$ sec. and plot them in Fig. 14.

As shown in Fig. 14, lower damping is associated with higher variance in the estimate, which was analytically derived in Equation (21). Naturally, the biggest effect is on $\widehat{a}_2$, the estimate of the 2 Hz component of the clock error, since the system mode is also present at that frequency. For low damping cases, the results can be improved by using a longer estimation window. To demonstrate that, we use a longer estimation window of $T = 60$ sec., with the results plotted below (Fig. 15), which show reduced variance on the estimate, as expected. However, it is important to note that in cases where the clock error is time varying (say the effect of thermal stress [28]), using a longer window results in a loss of time resolution for error tracking.

5.3. Comparison to time skew estimation approach [9]

The topic of periodic clock errors was first presented in [9], with an estimation approach to correct the phase angles, which as we know are hit the hardest. The corrupted time stamps are assumed to drift linearly w.r.t the true time, i.e., $t = \tau + (c_0 + c_1 \tau)$. Under periodic correction, this error resets to 0 on top of every second, which can be modeled as $t - \tau = \Delta t(\tau) = c_1(\tau - \lfloor \tau \rfloor)$ where $\lfloor * \rfloor$ is the greatest integer function. To estimate a correction for the corrupted phase angle $y_\psi(\tau)$, the proposed approach requires error-free angle measurements. Let us denote it by $y^{ref}(\tau)$. The clock error will result in a relative, sub-second drift between the two angles followed by a reset, which can be modeled as,

$$y^{ref}(\tau) - y_\psi(\tau) = \beta_0 + \beta_1(\tau - \lfloor \tau \rfloor)$$

Here it is assumed that the two angles have the same underlying

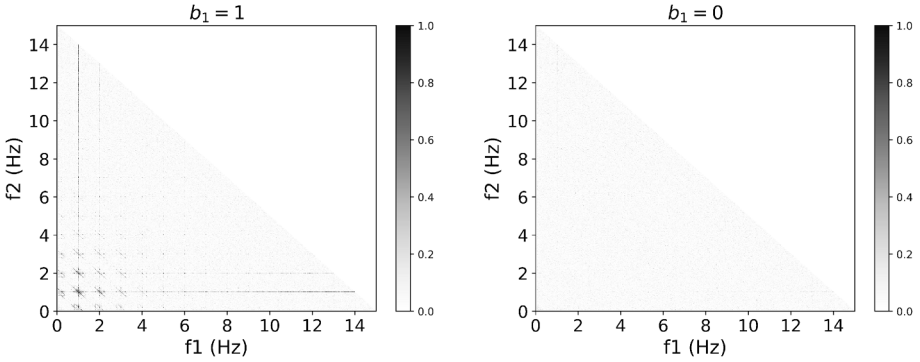


Fig. 10. Effect of Trend on Bicoherence of Clock-Corrupted AR Process.

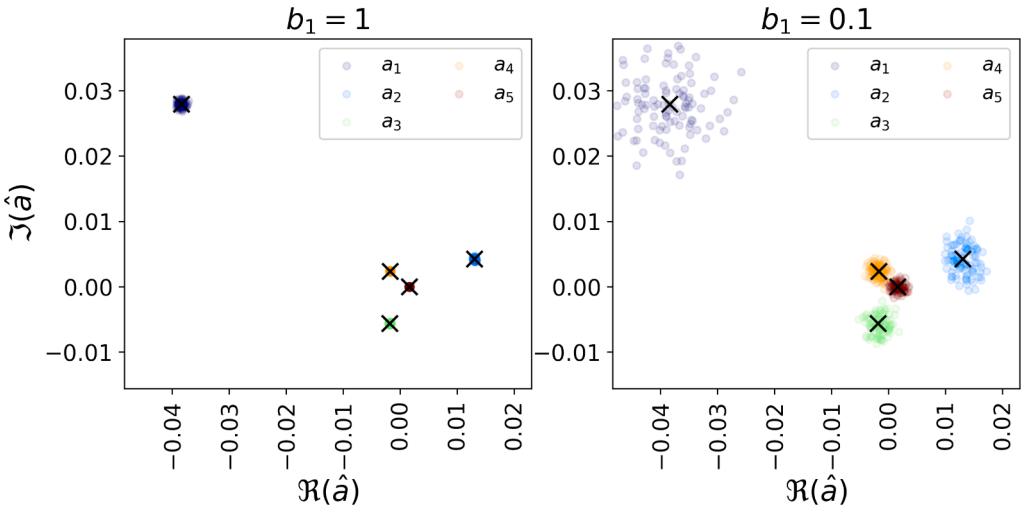


Fig. 11. Effect of Trend on Clock Error Estimate.

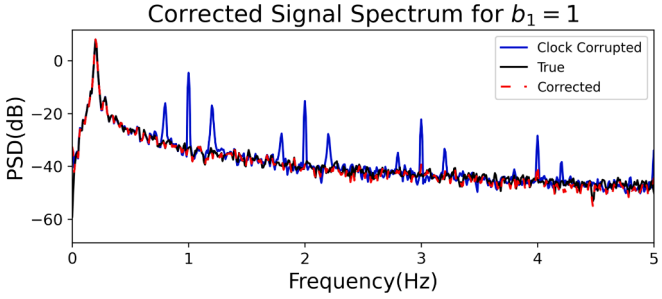


Fig. 12. Example 3 Dynamics with Corrected Clock.

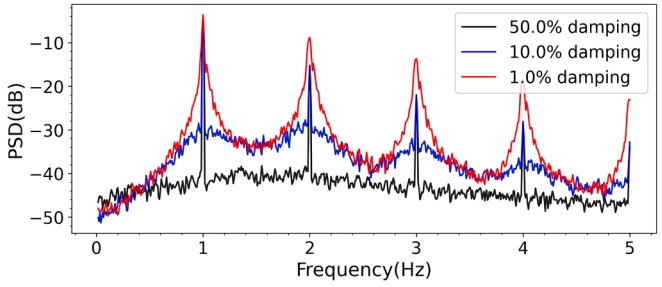


Fig. 13. Clock-Corrupted Signal PSD for Mode at 2 Hz.

frequency (no relative drift due to system dynamics), which holds in general in measurements from the same system. However, they can have different steady state angles (operating point), which is captured by the β_0. The above problem is solved iteratively through linear regression using a Kalman Filter, which allows to track the changing properties of clock errors if present. Note that it is also possible to batch estimate β_0, β_1 using least squares in the case of constant parameters, which holds in general. There are major differences between this approach and ours. Firstly, no theoretical insight into the statistical properties of their proposed estimator were provided in [9], particularly the role of stability of ambient dynamics in the estimator's performance. More importantly, the approach in [9] requires a reference angle free from errors, which as we have discussed previously, is not always possible to provide in practice. In addition, a periodic clock error model of the form $\Delta t(\tau) = c_1(\tau - \lfloor \tau \rfloor)$ is assumed in [9], while the model used in this manuscript is a generic periodic signal $\Delta t(\tau) = \sum_k a_k e^{j2\pi k f_0 t}$. As mentioned at the end of Section 4.2.1, we can easily adopt any other model, and in the case of the model used by [9], the unidentifiable a_0 can be estimated as $\hat{a}_0 = -\sum_k \hat{a}_k$, which we will use in this section to provide a fair comparison between the two methods. Finally, it is important to emphasize that the effect of clock errors is not only noticeable in phase angles, but in all types of signals from PMUs (at varying degrees) and therefore, instead of posing it as a phase angle correction problem, the goal should be to estimate the corrected time stamp, which can then be used to correct for all other signals from a PMU. In this section, we will mainly compare with the work in [9] on the first point, i.e. whether a reference is truly required.

For analysis, we create the two-phase angles $y_\psi(\tau)$ and $y^{ref}(\tau)$, having a 60 Hz fundamental frequency along with ambient dynamics from a system with two modes of frequencies 0.6 Hz and 1.5 Hz, respectively. The tiny periodic clock error is modeled as $\Delta t(\tau) = 10^{-5} \times (\tau - \lfloor \tau \rfloor)$ and results in a maximum of 0.2 degrees error in the phase angle as shown in Fig. 16, which easily meets the standard [5].

Fig. 17 shows the spectrum of the relative angle (w.r.t. the reference angle), i.e. $y^{ref}(\tau) - y_\psi(\tau)$, for both true and clock corrupted angle. The effect of periodic correction is clearly visible in the form of spectral spikes at 1 Hz and its multiples. Note that the signal power in ambient dynamics is in the range similar to what we saw in the actual observations in Fig. 4, to show how error standards were not made considering ambient data in mind. Now, in practice, a reference angle would be at a location different from the one being corrected and therefore, there will be a discrepancy in their ambient dynamics. This was emulated in this case too where the relative angle spectrum is not zero but has the two modes visible 0.6 and 1.5 Hz. It is important to remember that when using a reference, we must ensure that the discrepancy in system dynamics does not leak into the angle correction algorithm. Finally, we estimate the clock error using our proposed approach that doesn't need a reference and compare it against the estimate from [9]. The "corrected" phase angle spectra are plotted in Fig. 18, where not only is our estimate as good in accuracy, but it also questions the importance of reference angles in correcting periodic clock errors.

5.4. Real world measurements

Finally, we obtain the results for real-world synchrophasor data corresponding to Figs. 3 and 5 applying the proposed algorithm. First, the spectrogram from Fig. 3 is magnified in the low frequency range for clarity and shown in Fig. 19. A few prominent time–frequency traces are also labeled for discussion, where trace 1 is the clock mode itself, as analyzed next.

To detect the clock error, we take a 10 min window of ambient phase

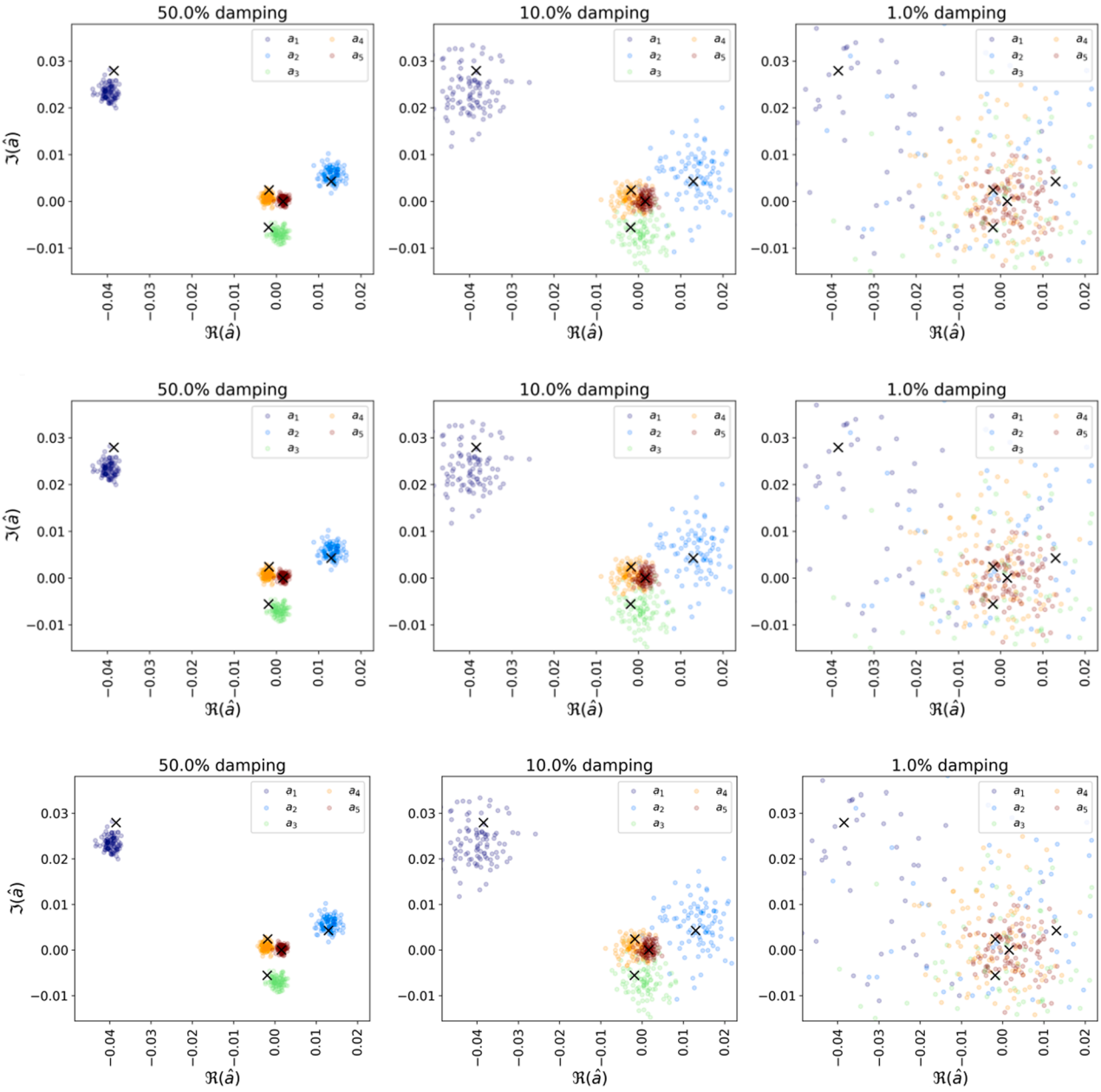


Fig. 14. Clock Error Estimates for Different Damping with $T = 10s$.

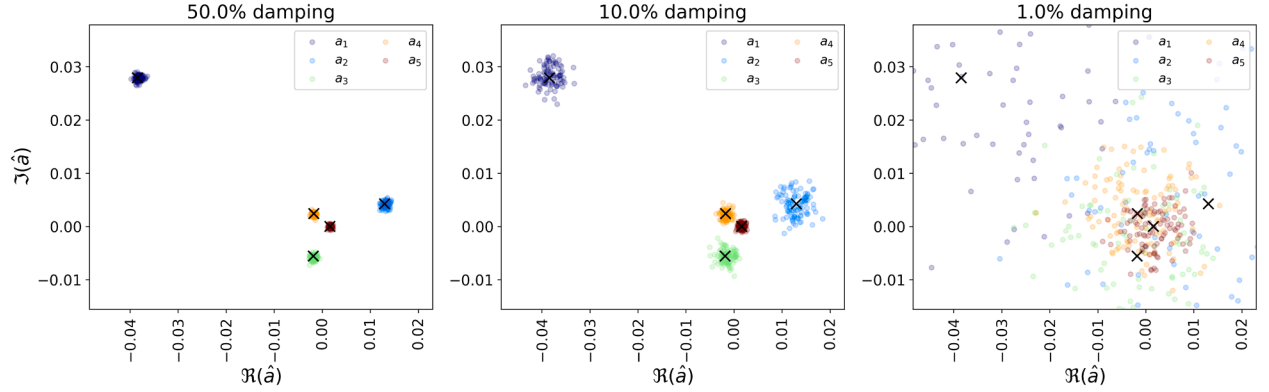


Fig. 15. Clock Error Estimates with Longer Window T = 60 s.

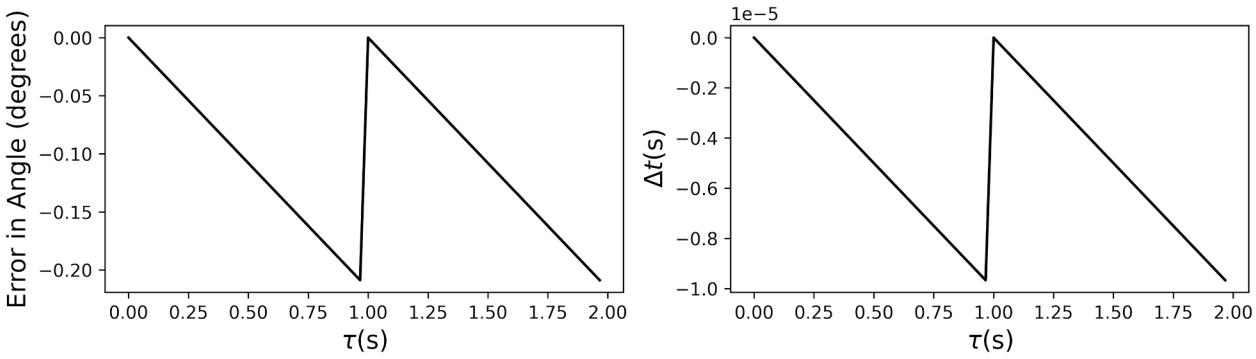


Fig. 16. Periodic Clock Error and its Resulting Phase Angle Error.

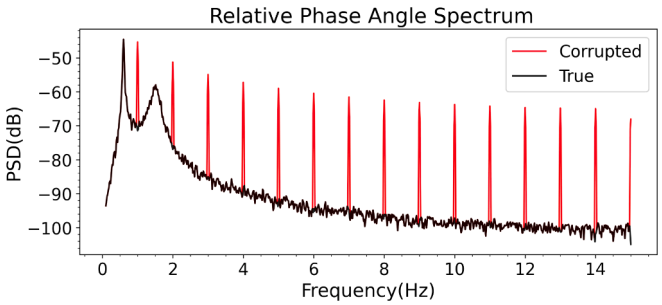


Fig. 17. Phase Angle Spectrum Relative to Reference Angle.

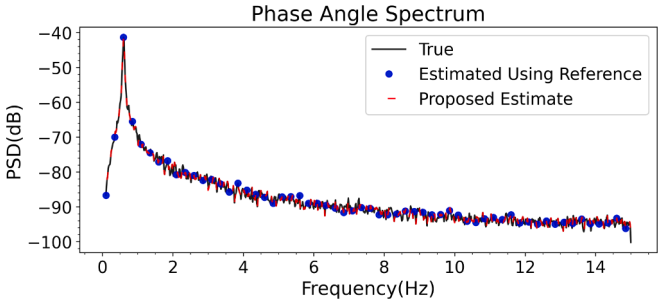


Fig. 18. Comparison of Phase Angle Correction to [9].

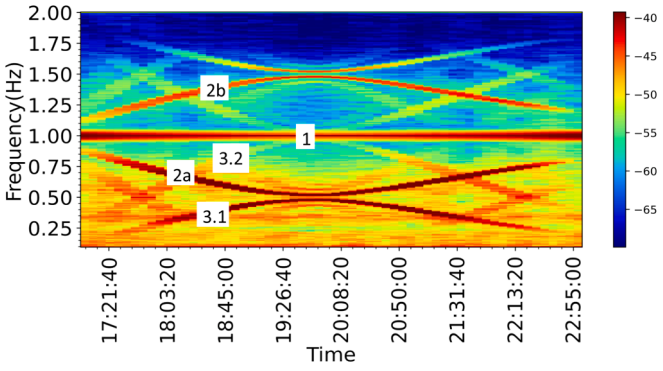


Fig. 19. Spectrogram Fig. 1 <2 Hz.

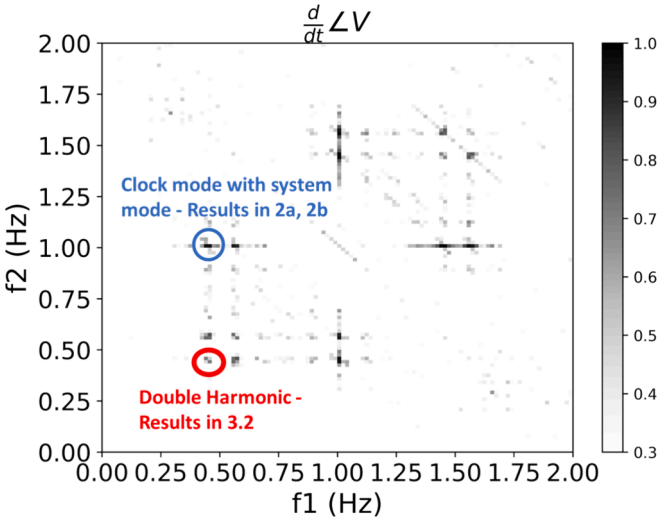


Fig. 20. Bicoherence for Fig. 3.

angle data from 19:00–19:10, preprocess it by taking a derivative to make it mean stationary and estimate its Bicoherence as shown in Fig. 20. High Bicoherence is seen between a frequency slightly lower than 0.5 Hz and the clock mode at 1 Hz, which results in traces 2a and 2b in the spectrogram in Fig. 19 (trace 1). This points to a possible system mode at slightly lower than 0.5 Hz, given by the trace labeled 3.1. However, we also see a significant Bicoherence of this mode with itself, which points to a potential second harmonic, resulting in trace 3.2. The remaining Bicoherence results can be explained by analysing Table 1, assuming a mode at ~0.5 Hz, 2 x 0.5 Hz and clock frequencies at 1 Hz and its multiples.

Finally, we estimate the clock error using the proposed algorithm and plot it in in Fig. 21 along with the corrected phase angle. The clock error is low yet able to create a significant effect on the phase angle spectrum as seen before. From the bottom subplot, one can see that the corrected phase angle is relatively smoother as expected compared to the measured phase angle, which has periodic sharp changes because of the abrupt clock corrections.

To provide additional real-world cases from the Dominion Energy system, we pick phase angle data from other power plants, labelled Sub 0, 1, 2 and 3 with their phase angle spectra plotted in Fig. 22. Firstly, one can notice discrepancies among the locations, which is expected. More importantly, all of them have prominent spectral spikes from periodic clock errors. One location, which is a combined cycle plant labelled Sub 2, has a genuine, poorly damped mode at 10 Hz, which could pose a challenge to clock error estimation. The result of time stamp error estimation and the resulting phase angle correction is shown in Fig. 23. The correction levels are in the order of microseconds, which is what makes the data acceptable the standards point of view. However, we can clearly see the level of corruption in the frequency domain, where the resulting spectral spikes clearly dominate the information. Also, notice the estimated correction for substation 2 in Fig. 23. The poorly damped dynamics at 10 Hz have leaked into the clock correction, as discussed in

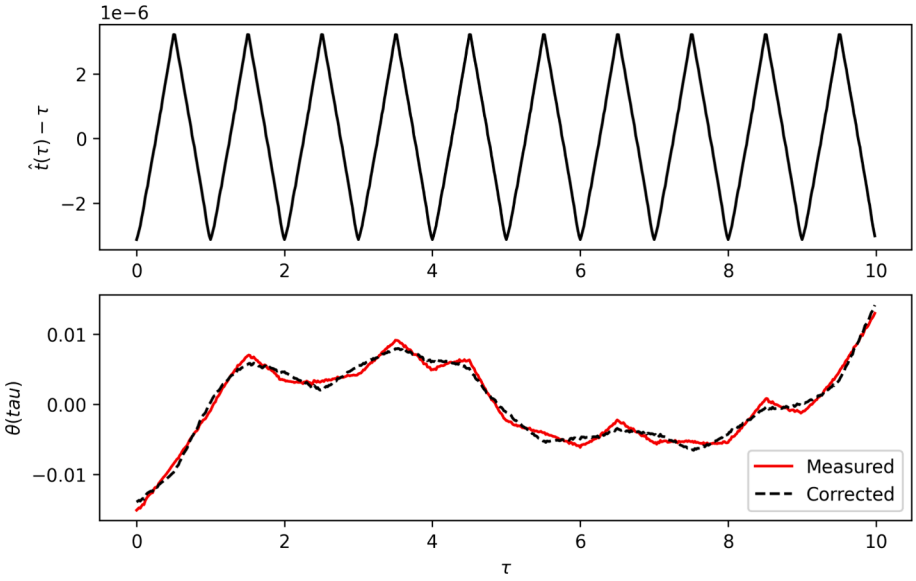


Fig. 21. Estimated Correction.

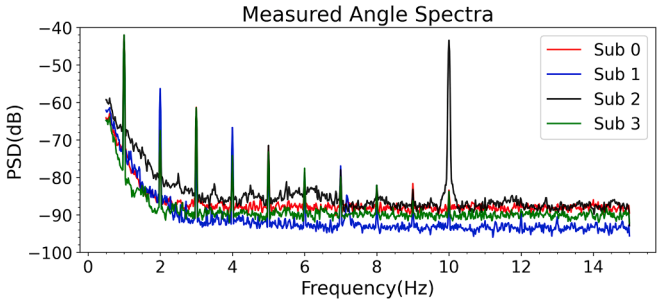


Fig. 22. Phase Angles at Other Locations in Dominion Energy System Showing Periodic Clock Effects.

previous sections, thereby biasing the estimate. Here, note that all 4 synchrophasor devices belong to the same vendor and more importantly, have identical configuration. In such a scenario, one could explore using clock correction results from one to fix the other, to circumvent location specific issues, such as the undamped mode in Sub 2. Finally, to confirm quality of clock correction, we plot the corrected phase angle spectra in Fig. 24, which shows how the spectral spikes from periodic clock errors have been removed while preserving essential power system dynamics.

6. Conclusion and future work

This work presented a detailed analysis of how periodic clock errors and drift can affect measurements, particularly those of phase angles, to the extent that they are no longer fit for use in ambient synchrophasor analytics. The Bicoherence's ability to distinguish between clock modes and real system dynamics has also been shown. Furthermore, an estimator is proposed to recover the true time stamps (partially) from ambient measurements without a prior knowledge of true system dynamics.

While the proposed approach shows promise, there improvements needed to address real world issues arising from internal clock errors, which is expected to pave the way for future research in this domain:

1. In practice, the clock errors can manifest in phase angles in the form of drifts (due to lack of synchronization between clocks of different devices) and periodic components due to correction as shown in Figs. 1 and 2. To address them together, there is a need to be able to automatically identify and quantify the impact of each type of error and to correct for them accordingly. This would require an approach working with PMU data from multiple locations.
2. The proposed approach for estimating errors can be affected by stable power system (mostly local) dynamics, in which case, the clock correction estimates will become biased. In this case, having an "error-free" reference angle could be helpful. However, a more detailed investigation on the added properties the reference angle dynamics should satisfy is needed. Alternatively, using similarities in synchrophasor devices from the same vendor could be exploited to use estimates from one location to correct the other.
3. The presented internal clock errors can heavily impact the downstream applications, particularly the ones around power system oscillation monitoring since these periodic components do not follow regular system dynamics. Leaving this issues unaddressed would result in false alarms or not being able to detect major system dynamic changes, similar to the effect of forced oscillations in [29]. Consequently, there is a need to adapt the existing mode estimation approaches to account for clock errors. For example, adding terms in mode estimation methods could help eliminate the periodic error's effects, similar to the work to deal with forced oscillations in [30].

CRediT authorship contribution statement

Chetan Mishra: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Luigi Vanfretti:** Writing – review & editing, Supervision, Methodology, Investigation, Conceptualization. **Jaime Delaree:** Writing – review & editing, Project administration. **Kevin D. Jones:** Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

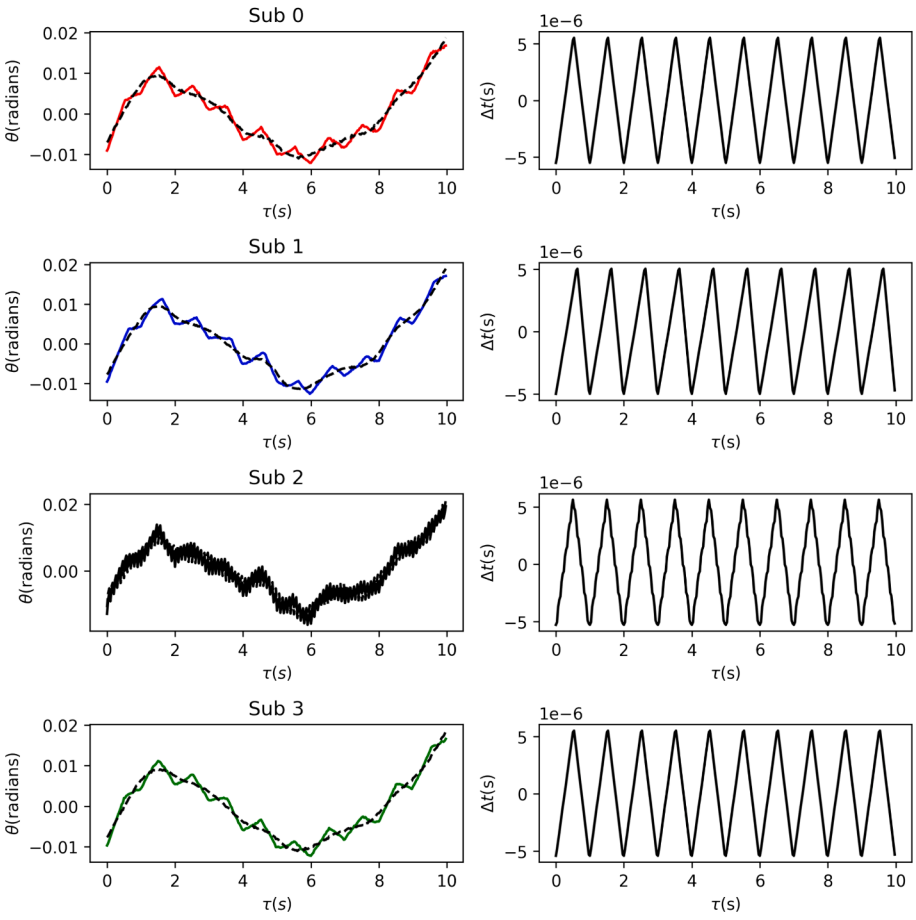


Fig. 23. Left: Phase Angle Correction (black dotted) with Measured (colored). Right: Time Stamp Correction.

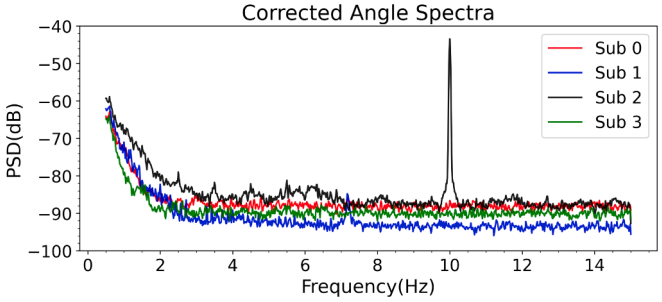


Fig. 24. Corrected Phase Angle Spectra.

Appendix A. Statistical properties of $y_{\psi}^{lin}(\tau)$

In this section, define $y_{\psi}^{lin}(\tau) = \int h(\tau,\eta)u(\eta)d\eta = \int (g(\tau-\eta)+\partial_{\tau}g(\tau-\eta)\Delta t(\tau))u(\eta)d\eta = \int g(\eta)\{u(\tau-\eta)+\dot{u}(\tau-\eta)\Delta t(\tau)\}d\eta$. Before proceeding, we derive the a few statistical properties of $y_{\psi}^{lin}(\tau)$ for zero mean white noise $u(t)$ i.e. $E(u(t)) = 0, E(u(t)u(t_1)) = \delta(t-t_1)$.

$$E(y_{\psi}^{lin}(\tau)) = \int h(\tau,\eta)E(u(\eta))d\eta = 0 \tag{23}$$

$$E\left(y_{\psi}^{lin}(\tau)y_{\psi}^{lin}(\tau_1)\right) = \int h(\tau,\eta)h(\tau_1,\eta)d\eta = \underbrace{\left(\int g(\tau-\eta)g(\tau_1-\eta)d\eta\right)}_{R(\tau-\tau_1)} + \underbrace{\left(-\int g(\tau-\eta)\partial_\eta g(\tau_1-\eta)d\eta\right)}_{R_{11}(\tau-\tau_1)}\Delta t(\tau_1)$$

$$+\underbrace{\left(-\int g(\tau_1-\eta)\partial_\eta g(\tau-\eta)d\eta\right)}_{R_{12}(\tau-\tau_1)}\Delta t(\tau) + \underbrace{\left(\int \partial_\eta g(\tau-\eta)\partial_\eta g(\tau_1-\eta)d\eta\right)}_{R_2(\tau-\tau_1)}\Delta t(\tau)\Delta t(\tau_1)$$

Appendix B. Proofs of lemmas

B.1 Lemma 1

$$\underset{T\to\infty}{\text{Lim}}\frac{\int_0^T y_{\psi}^{lin}(\tau)d\tau}{T} = \int g(\eta)\int_0^T \frac{u(\tau-\eta)+\dot{u}(\tau-\eta)\Delta t(\tau)}{T}d\tau d\eta = \int g(\eta)\int_0^T \frac{u(\tau-\eta)}{T}d\tau + \lim_{N\to\infty}\int_0^{\frac{2\pi}{\omega_0}}\sum\nolimits_{n=0:N-1}\frac{\dot{u}\left(\frac{2\pi n}{\omega_0}+\tau-\eta\right)}{\frac{N2\pi}{\omega_0}}\Delta t(\tau)d\tau d\eta$$

$$\xrightarrow{p} \int g(\eta)\left(E(u)+E(\dot{u})\int_0^{\frac{2\pi}{\omega_0}}\Delta t(\tau)d\tau\right)d\eta = 0 \text{(weak law of large numbers and Slutsky's theorem). Consequently,}$$

$$\underset{T\to\infty}{\text{Lim}}\frac{\int_0^T e(\tau)d\tau}{T^2} = \underset{T\to\infty}{\text{Lim}}\frac{\int_0^T b_1\Delta t(\tau)d\tau}{T^2} + \frac{\int_0^T y_{\psi}^{lin}(\tau)d\tau}{T^2} \xrightarrow{p} \frac{b_1 a_0}{T} = 0 \tag{25}$$

B.2 Lemma 2

$$\frac{1}{T^3}E\left(\int_0^T \tau e(\tau)d\tau\right) = \frac{1}{T^3}E\left(\int_0^T \tau\left(b_1\Delta t(\tau)+y_{\psi}^{lin}(\tau)\right)d\tau\right) = \frac{1}{T^3}\int_0^T \tau b_1\Delta t(\tau)d\tau$$

$$VAR\left(\frac{\int_0^T \tau e(\tau)d\tau}{T^3}\right) = VAR\left(\frac{\int_0^T \tau y_{\psi}^{lin}(\tau)d\tau}{T^3}\right) = \frac{1}{T^6}\int_0^T\int_0^T \tau\tau_1 E\left(y_{\psi}^{lin}(\tau)y_{\psi}^{lin}(\tau_1)\right)d\tau d\tau_1$$

Here, the expression for $E\left(y_{\psi}^{lin}(\tau)y_{\psi}^{lin}(\tau_1)\right)$ is given in (23). For stable systems $R(t), R_{11}(t), R_{12}(t)$ and $R_2(t)$ exponentially decay with t. Consequently, $\lim_{T\to\infty}\int_0^T \tau_1 E\left(y_{\psi}^{lin}(\tau)y_{\psi}^{lin}(\tau_1)\right)d\tau_1 < \infty$. Therefore, $\lim_{T\to\infty} VAR\left(\frac{\int_0^T \tau e(\tau)d\tau}{T^3}\right)\to 0$ and by Chebyshev's inequality [17], yielding

$$\lim_{T\to\infty}\frac{\int_0^T \tau e(\tau)d\tau}{T^3} \xrightarrow{p} 0 \tag{26}$$

Appendix C. . Properties of clock error estimator $\widehat{a}_k$

$$E(\widehat{a}_k|\widehat{b}_1 = b_1) = \frac{1}{b_1 T}\int_0^T \left(b_0 + b_1\Delta t(\tau) + \int h(\tau,\eta)\underbrace{E(u(\eta))}_{=0}d\eta\right)e^{-j2\pi k f_0\tau}d\tau = \frac{1}{b_1 T}\int_0^T (b_0 + b_1\Delta t(\tau))e^{-j2\pi k f_0\tau}d\tau = a_k$$

$$VAR(\widehat{a}_k(T)|\widehat{b}_1 = b_1) = \frac{1}{b_1^2 T^2}\int_0^T\int_0^T E\left(y_{\psi}^{lin}(\tau)y_{\psi}^{lin}(\tau_1)\right)e^{-j2\pi k f_0(\tau-\tau_1)}d\tau d\tau_1$$

For small clock errors $\Delta t(\tau)$, only retaining the first term in (23) yields,

$$VAR(\widehat{a}_k(T)|\widehat{b}_1 = b_1) \approx \frac{1}{b_1^2}\int_0^T\int_0^T \frac{1}{T^2}R(\tau-\tau_1)e^{-jk2\pi f_0(\tau-\tau_1)}d\tau d\tau_1$$

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